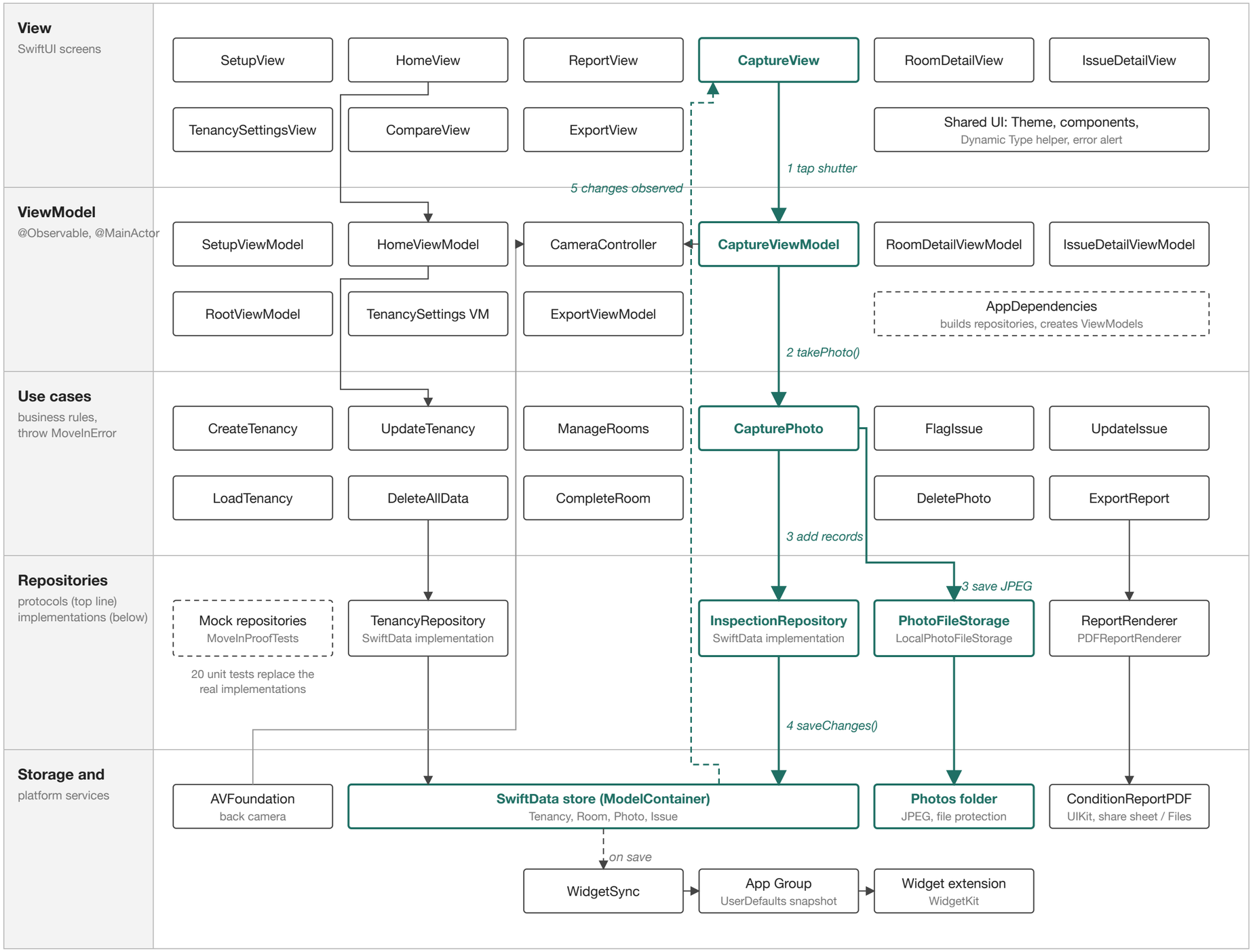
MoveIn Proof: architecture and data flow

MVVM + use cases + repositories. SwiftUI, SwiftData, AVFoundation, WidgetKit. All storage is on the device.



Taking a photo

1. CaptureView sends the tap to CaptureViewModel.
2. The ViewModel captures the image with CameraController, stamps the date and time on it and calls CapturePhoto.
3. CapturePhoto saves the JPEG through PhotoFileStorage and adds the Photo (and an Issue if tagged) through InspectionRepository.
4. saveChanges() commits to SwiftData. If that fails the JPEG is deleted and MoveInError.storageFailed is thrown.
5. SwiftData observation updates the screen. Errors are shown as an alert by the ViewModel. WidgetSync also writes a snapshot for the widget.

Data model

Deleting a room cascades to its photos and issues.
Deleting an issue keeps its photos.

Notes

Each layer only calls the one below it.
Views never import SwiftData; use cases only see repository protocols.
UI state runs on the main actor; the camera session has its own serial queue.
No network access. The widget reads a copy of the key dates from the App Group.

Key

- calls / depends on
- path traced above
- notification / observation
- wiring or test-only component